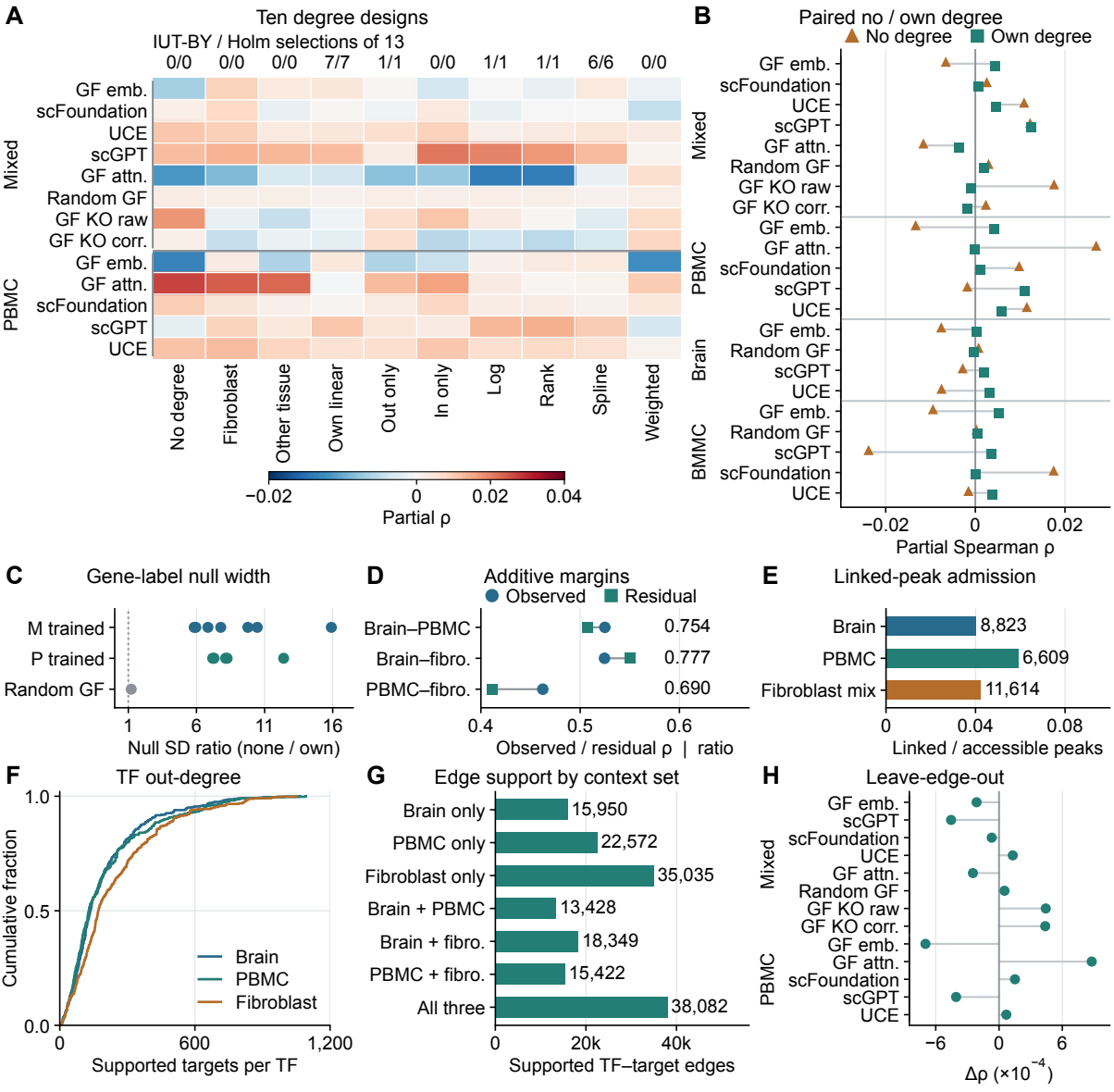
A
Ten degree designs
IUT-BY / Holm selections of 13
0/0 0/0 0/0 7/7 1/1 0/0 1/1 1/1 6/6 0/0
Mixed
PBMC
GF emb.
scFoundation
UCE
scGPT
GF attn.
Random GF
GF KO raw
GF KO corr.
GF emb.
GF attn.
scFoundation
scGPT
UCE
No degree
Fibroblast
Other tissue
Own linear
Out only
In only
Log
Rank
Spline
Weighted
−0.02
0
0.02
0.04
Partial ρ
B
Paired no / own degree
No degree
Own degree
Mixed
PBMC
Brain
BMMC
GF emb.
scFoundation
UCE
scGPT
GF attn.
Random GF
GF KO raw
GF KO corr.
GF emb.
GF attn.
scFoundation
scGPT
UCE
GF emb.
Random GF
scGPT
UCE
GF emb.
Random GF
scGPT
scFoundation
UCE
−0.02
0
0.02
Partial Spearman ρ
C
Gene-label null width
M trained
P trained
Random GF
1
6
11
16
Null SD ratio (none / own)
D
Additive margins
Observed
Residual
Brain–PBMC
Brain–fibro.
PBMC–fibro.
0.754
0.777
0.690
0.4
0.5
0.6
Observed / residual ρ | ratio
E
Linked-peak admission
Brain
PBMC
Fibroblast mix
8,823
6,609
11,614
0
0.04
0.08
Linked / accessible peaks
F
TF out-degree
1.0
0.5
0.0
Cumulative fraction
Brain
PBMC
Fibroblast
0
600
1,200
Supported targets per TF
G
Edge support by context set
Brain only
PBMC only
Fibroblast only
Brain + PBMC
Brain + fibro.
PBMC + fibro.
All three
15,950
22,572
35,035
13,428
18,349
15,422
38,082
0
20k
40k
Supported TF–target edges
H
Leave-edge-out
Mixed
PBMC
GF emb.
scGPT
scFoundation
UCE
GF attn.
Random GF
GF KO raw
GF KO corr.
GF emb.
GF attn.
scFoundation
scGPT
UCE
−6
0
6
Δρ (×10⁻⁴)